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COURSE TITLE

Microprocessors and Microcontrollers

COURSE CODE

BECE204L

Course Project Report

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Topic: Microcontroller-Based Digital Door Lock with Security Alert

1. Background Summary

With the advancement of embedded systems and automation technologies, there is a growing need for *reliable and secure access control systems* in residential and commercial environments. Traditional mechanical locks are prone to issues such as key duplication, loss of keys, and lack of user feedback, making them less secure.

To overcome these limitations, electronic locking systems have been developed using microcontrollers and peripheral devices. This project presents a smart door lock system that uses a keypad-based password authentication mechanism. The system integrates a solenoid lock controlled via a relay module, an OLED display for user interaction, LEDs for status indication, and a buzzer for audio alerts.

The system enhances security by allowing only authorized access through password verification. It also incorporates a lockout mechanism after multiple failed attempts, thereby improving protection against unauthorized access. This project demonstrates the practical application of embedded systems in modern security solutions.

Table 1 : System performance metrics

Metric	Value	Notes
Password Verification Time	< 50 ms	Software comparison (string match)
Keypad Scan Rate	~100 Hz	Software polling
Total Authentication Time	< 3 s	Includes user input time
Failed Attempt Limit	4 attempts	Configured in software
Lockout Duration	30 s	After max failed attempts
Solenoid Unlock Duration	5 s	Configurable in code
Buzzer Response Time	< 100 ms	Immediate audio feedback
Display Response Time	< 200 ms	OLED update delay
Relay Switching Time	~10–20 ms	Hardware dependent
Operating Voltage	3.3V / 5V / 12V	MCU / Logic / Solenoid

2. Literature Report

Electronic security systems have evolved significantly with the integration of embedded systems and microcontrollers. Several studies have focused on replacing traditional locking mechanisms with digital authentication systems such as keypad-based locks, RFID-based access control, and biometric systems.

Keypad-based locking systems are widely used due to their simplicity, cost-effectiveness, and ease of implementation. Research shows that combining multiple feedback mechanisms such as visual displays and audio alerts improves user interaction and system reliability. OLED displays have gained popularity due to their low power consumption and high contrast, making them suitable for embedded applications.

Additionally, security systems often include fail-safe mechanisms such as lockout features to prevent brute-force attacks. The use of relays for controlling high-power devices like solenoid locks is also a standard practice in embedded system design.

This project builds upon these concepts by integrating password authentication, real-time feedback using an OLED display, and a security lockout feature, providing a comprehensive and efficient smart locking solution.

3. Working Principle

The system operates based on password authentication using a keypad interface. When powered on, the microcontroller initializes all components including the OLED display, relay module, LEDs, and buzzer.

The user enters a password through the keypad, which is captured and stored temporarily. The entered password is compared with a predefined correct password stored in the system. If the password matches, the relay is activated, which in turn energizes the solenoid lock to unlock the door. Simultaneously, a green LED is turned on and a success tone is generated using the buzzer.

If the entered password is incorrect, the system increments a failed attempt counter and provides feedback through a red LED and a warning sound. The OLED display shows appropriate messages such as “Wrong Password” and remaining attempts.

If the number of failed attempts exceeds the maximum limit, the system enters a lockout mode for 30 seconds, during which access is denied. After the lockout period, the system resets and allows the user to attempt again.

The entire operation is controlled through programmed logic implemented in the microcontroller, ensuring reliable and secure functioning.

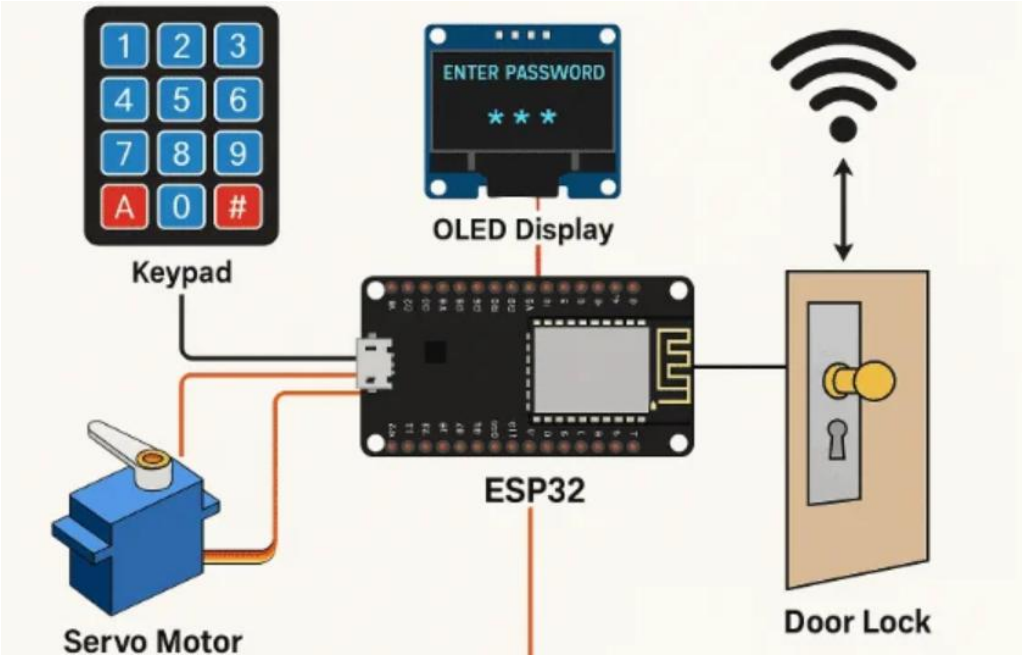


Image 1: Overall idea of door-lock

4. Experimental Validation

The proposed system was implemented using a microcontroller, keypad, OLED display, relay module, solenoid lock, LEDs, and a buzzer. The circuit was assembled on a breadboard using jumper wires, and the system was powered using a regulated power supply. Multiple test cases were conducted to validate the system performance. When the correct password was entered, the solenoid lock successfully activated, unlocking the door for a duration of 5 seconds as programmed. The OLED display showed “Access Granted,” and the buzzer produced a success tone. When incorrect passwords were entered, the system displayed error messages and activated the red LED along with a warning tone. After four consecutive incorrect attempts, the system entered a lockout mode for 30 seconds, during which input from the keypad was ignored and a countdown timer was displayed. The system was tested repeatedly under different conditions, and consistent performance was observed. All components functioned as expected, validating the reliability of the design.

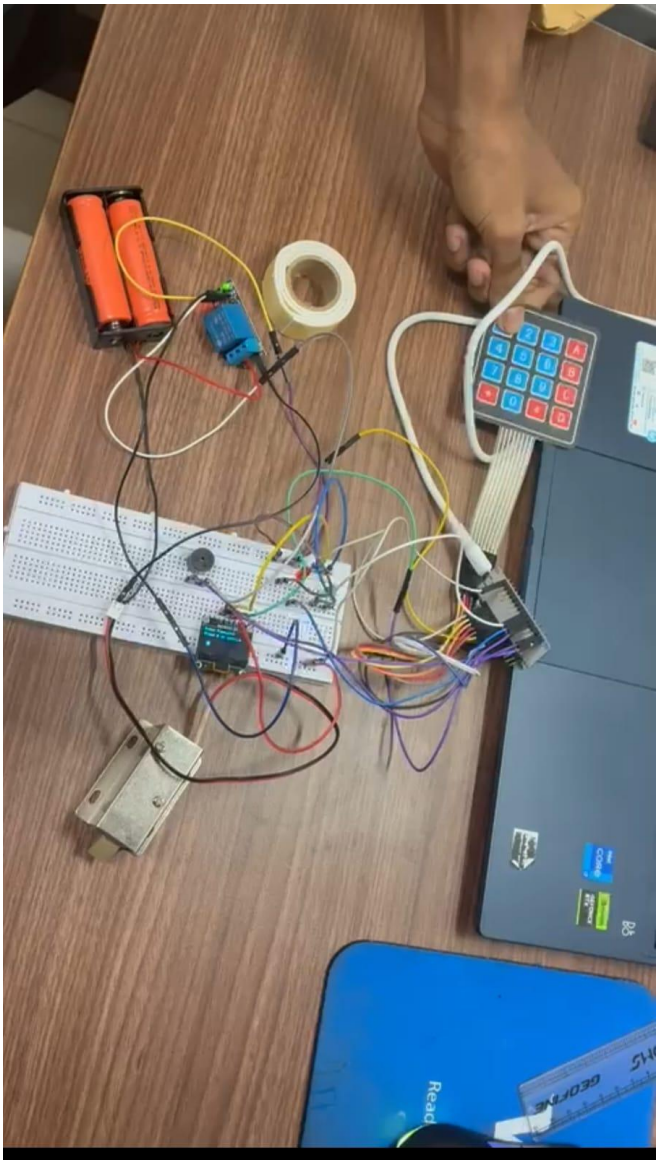
5. Results and Discussion

The implemented smart lock system successfully demonstrated secure and reliable access control using a keypad-based authentication mechanism. The integration of multiple components such as the OLED display, buzzer, LEDs, and relay module provided an interactive and user-friendly experience.

The system effectively granted access only when the correct password was entered and denied access for incorrect attempts. The lockout feature proved to be an important security enhancement, preventing repeated unauthorized attempts. The OLED display improved usability by providing real-time feedback to the user.

One of the key advantages of the system is its simplicity and cost-effectiveness, making it suitable for small-scale applications such as homes, hostels, and offices. However, the system can be further improved by incorporating advanced features such as biometric authentication, mobile app control, or IoT-based monitoring.

Overall, the project successfully meets its objectives and demonstrates the practical application of embedded systems in security solutions.



6. Inferences

From the experimental results and system performance analysis, it can be inferred that the proposed smart locking system operates efficiently and reliably for secure access control applications. The system successfully authenticates users based on a predefined password and provides immediate feedback through visual (OLED display and LEDs) and audio (buzzer) indicators.

The response time of the system is observed to be minimal, ensuring a smooth user experience. The implementation of a failed attempt counter and lockout mechanism significantly enhances security by preventing repeated unauthorized access attempts. The relay-controlled solenoid lock functions effectively, enabling and disabling access as intended.

Furthermore, the integration of multiple components demonstrates the effectiveness of embedded systems in real-world applications. The system is cost-effective, easy to implement, and suitable for small-scale security applications such as homes, hostels, and offices.

However, the system can be further improved by incorporating advanced features such as biometric authentication, remote access via IoT, or encryption techniques for enhanced security. Overall, the project meets its objectives and validates the feasibility of a microcontroller-based smart locking system.

7. SDG (Sustainable Development Goals)

This project aligns with several United Nations Sustainable Development Goals (SDGs), particularly SDG 9 (*Industry, Innovation and Infrastructure*), SDG 11 (*Sustainable Cities and Communities*), and SDG 16 (*Peace, Justice and Strong Institutions*). The development of a smart solenoid-based locking system demonstrates innovation in embedded systems and promotes the use of modern technology in building secure infrastructure. By providing an efficient and affordable access control solution, the project contributes to enhancing safety in residential and commercial environments, thereby supporting the creation of secure and sustainable communities. Furthermore, the implementation of password authentication and a lockout mechanism strengthens access control, helping to prevent unauthorized entry and improve overall security. Thus, the project reflects the practical application of technology in achieving safer, smarter, and more reliable living spaces.